

- 7 (a) e.g. storing energy
blocking d.c.
in oscillator circuits
in tuning circuits
in timing circuits

any two

B2 [2]

(b) (i) $1/6 + 1/C + 1/C = 1/4$

C1

$$C = 24\mu\text{F}$$

A1 [2]

(ii) $Q = CV$
 $= 4.0 \times 10^{-6} \times 12$

C1

$$= 48\mu\text{C}$$

A1 [2]

(iii) 1. $48\mu\text{C}$

A1

2. $24\mu\text{C}$

A1 [2]